Odinn Waguespack

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Education

Carnegie Mellon University — Pittsburgh, PA
M.S. Mechanical Engineering - Robotics and Controls — *May 2028*

University of Maryland — College Park, MD
B.S. Mechanical Engineering — *December 2024*

Experience

Carnegie Robotics — Pittsburgh, PA
Mechanical Engineering Intern — *May 2025 - Sep 2025*

- Assisted in the ground-up development of a fully self contained off-road autonomy platform
- Owned design, assembly, and testing of a perception payload (LiDAR, stereo cameras, IMU) on a Polaris RZR – modeled vehicle kinematics to optimize sensor field-of-view (FOV) coverage.
- Enabled drive-by-wire autonomy by retrofitting a custom electromechanical shift actuator and electronic power steering unit on vehicle.
- Performed thermal/flow-sim analysis of active ventilation, resulting in a 11% decrease to allocated power budget

National Institute Of Standards And Technology — Gaithersburg, MD
R&D Intern - Quantum Measurement Division — *Oct 2024 - April 2025*

- Developed a high precision torsion pendulum for integration with an atom interferometer testing the quantum coherence of gravity.
- Implemented python scripts to automate LabView data collection, enabling FFT-based spectral analysis of high-frequency noise.
- Designed an experimental ultra-high vacuum (UHV) test setup and managed part requisition, accelerating project timeline by 4 months.
- Ran root-cause failure analysis in CuBe/Si3N4 fibers: diagnosed inconsistencies in supplier manufacturing process; saving over $30,000 in build costs.

Terrapin Works - Machine Shop — College Park, MD
Lab Manager — *Dec 2022 - Dec 2024*

- Promoted to manager after 1.5 years as a machinist; managed all 5-axis CNC operations and equipment maintenance.
- Advised academic, industry, and government partners on design, manufacturing, and optimization across 25+ specialized projects.
- Led training of new machinists on CNC, waterjet, and lathe operations, emphasizing safe machine operation.

Agentis Air — Derwood, MD
Mechanical Engineering Intern — *May 2023 - August 2023*

- Researched a novel ionizing filtration system. Integrated prototype with existing commercial product, increasing system efficiency by 24%
- Wrote C++ scripts for real-time monitoring of Optical Particle Counter (OPC) sensor data, reducing individual test times by 30 minutes.

Projects and Activities

Robotic Manipulator, Senior Capstone (1st Place) — College Park, MD
Mechanical Design Lead — *Jan 2024 - May 2024*

- Led a team of five to design a 3-DOF robotic manipulator arm for integration with the Blue-ROV2 platform.
- Developed an SLA printed actuator for manipulator control, focusing on optimizing applied force and IP68 waterproof rating.
- Improved project cohesion by organizing design reviews to ensure technical specifications and deadlines were met.
- Placed 1st out of 37 competing projects.

Formula SAE — College Park, MD
Power Train Engineer — *Aug 2021 - Dec 2024*

- Designed a 4-1 exhaust manifold for optimization of scavenging, resulting in a 12% increase in peak torque at 10k rpm.
- Designed and CNC milled custom exhaust piping mount, using prototype FDM-printed scaffolding for welding.

MiniBot V2 — College Park, MD
Personal Project — *Aug 2025 - November 2026*

- Built a 4.4lb, 5-DOF bipedal robot capable of semi-autonomous terrain traversal.
- Redesigned electronics and compute mounting solutions for lowered COG, resulting in increased locomotion stability.
- Currently training improved walking policy in MuJoCo-based physics simulation environment.

Skills and Awards

Awards: Capstone Design Expo Deans Award, Deans List Fall 20-21, Spring 23-24
Programming: Python, C++, Matlab, MuJoCo, ROS2
Design/Software: Solidworks, Autodesk Fusion 360, Siemens NX, Excel, FEA, ANSYS, Creo, Overleaf/Latex
Manufacturing: Multi-axis CNC, Lathe, Manual Machining, 3D Printing (SLS, SLA, FDM, Powderbed), Laser Cutting, Waterjet